

Lecture 10: ML Training & Evaluation Finale

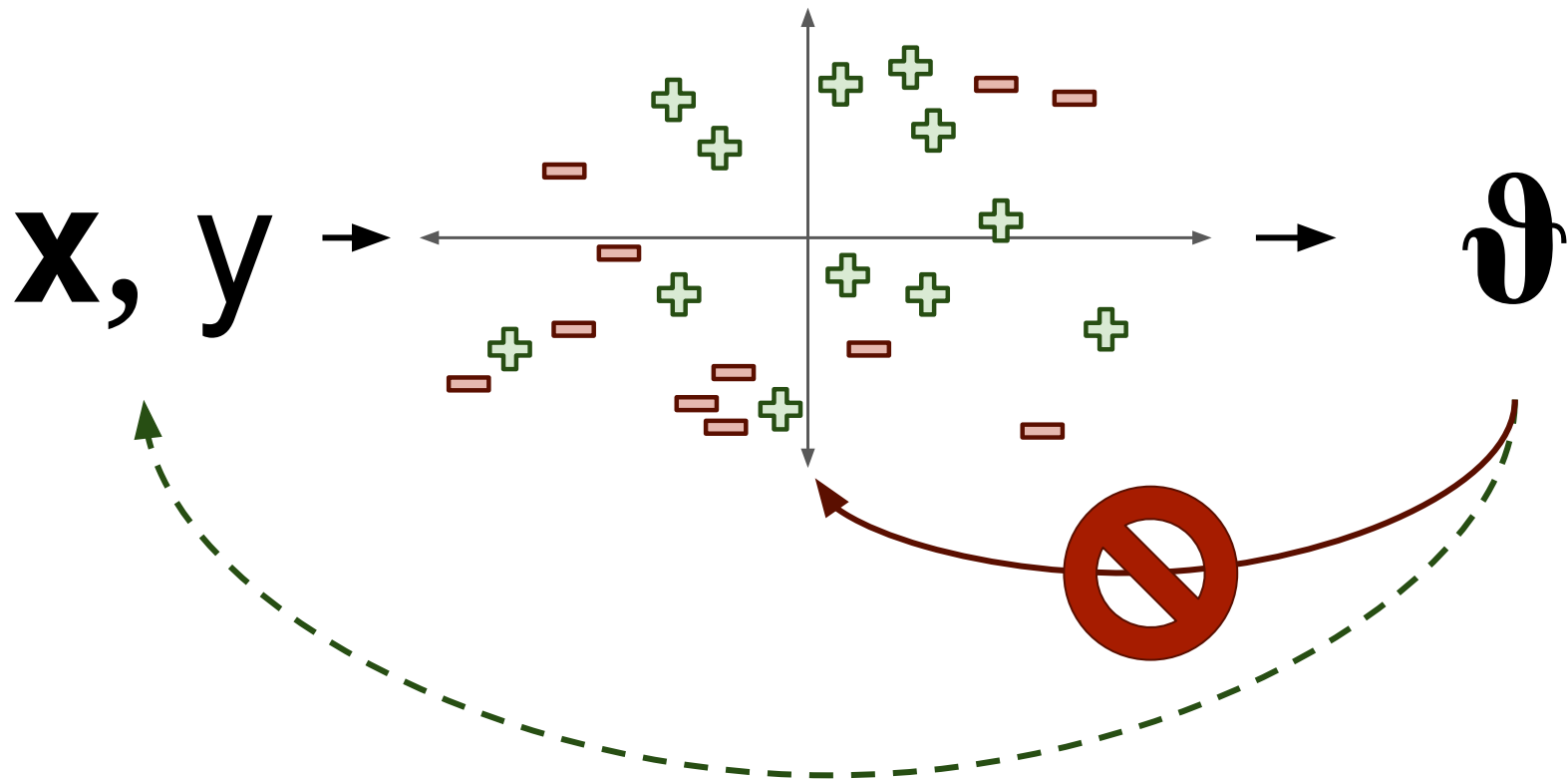
BINF 4002

Outline

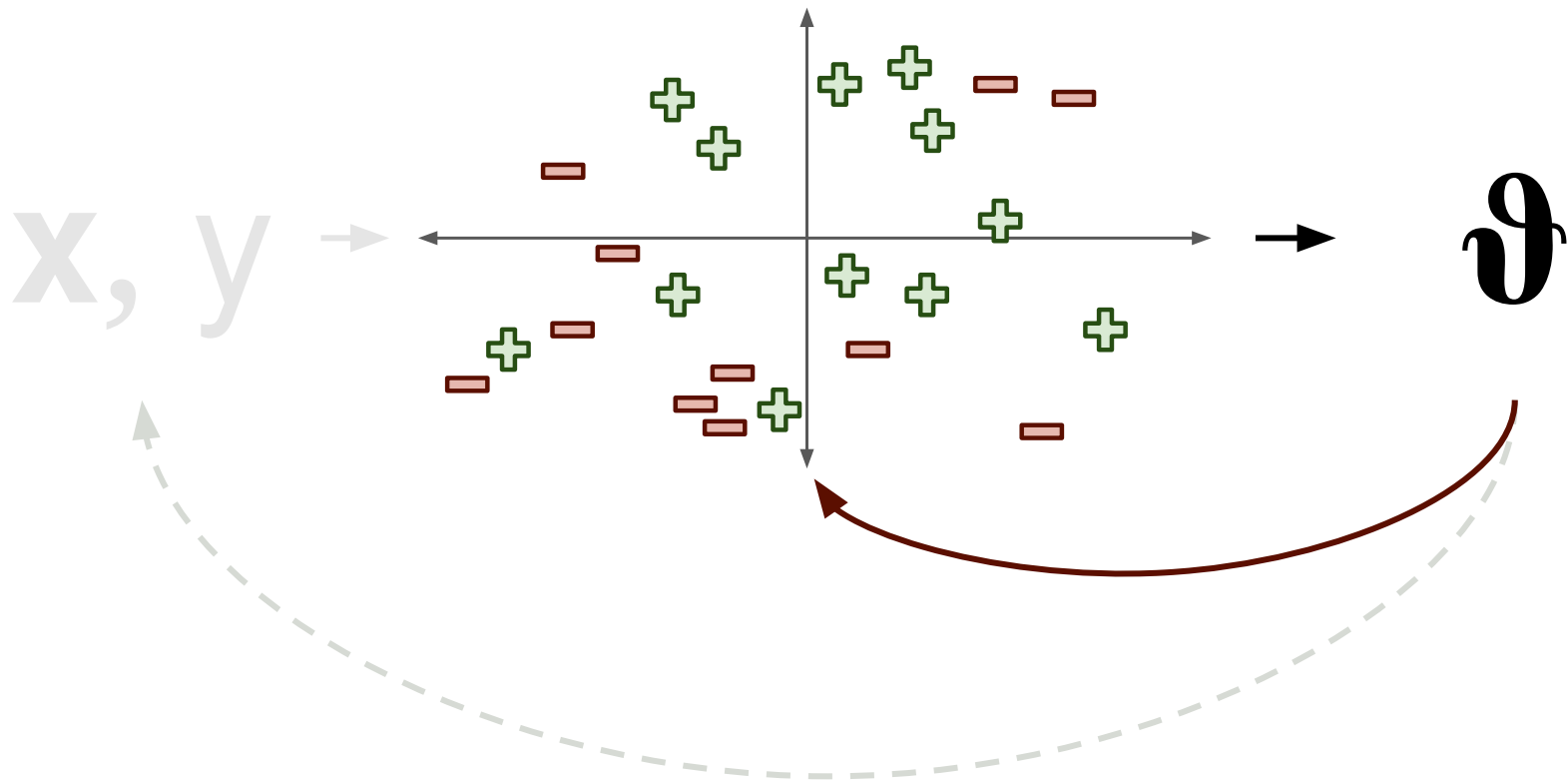
1. How (/ why) does ML actually work?
*Overfitting, memorization, and inductive bias**
2. How can this help you do ML better?
Guessing what will work and won't work and designing negative controls
3. When nothing (yet) will work.
Domain shift, Explainability, Fairness

* Zhang, C., Bengio, S., Hardt, M., Recht, B., & Vinyals, O. (2017). Understanding deep learning requires rethinking generalization. International Conference on Learning Representations. Retrieved from <https://openreview.net/forum?id=Sy8gdB9xx>

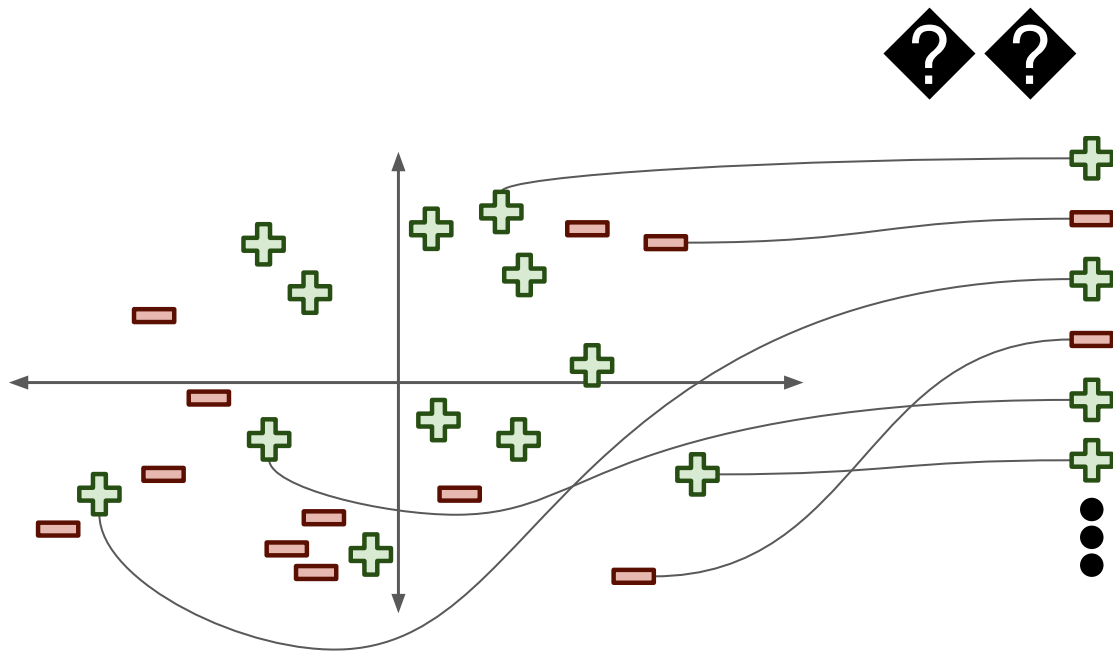
The Overfitting Problem



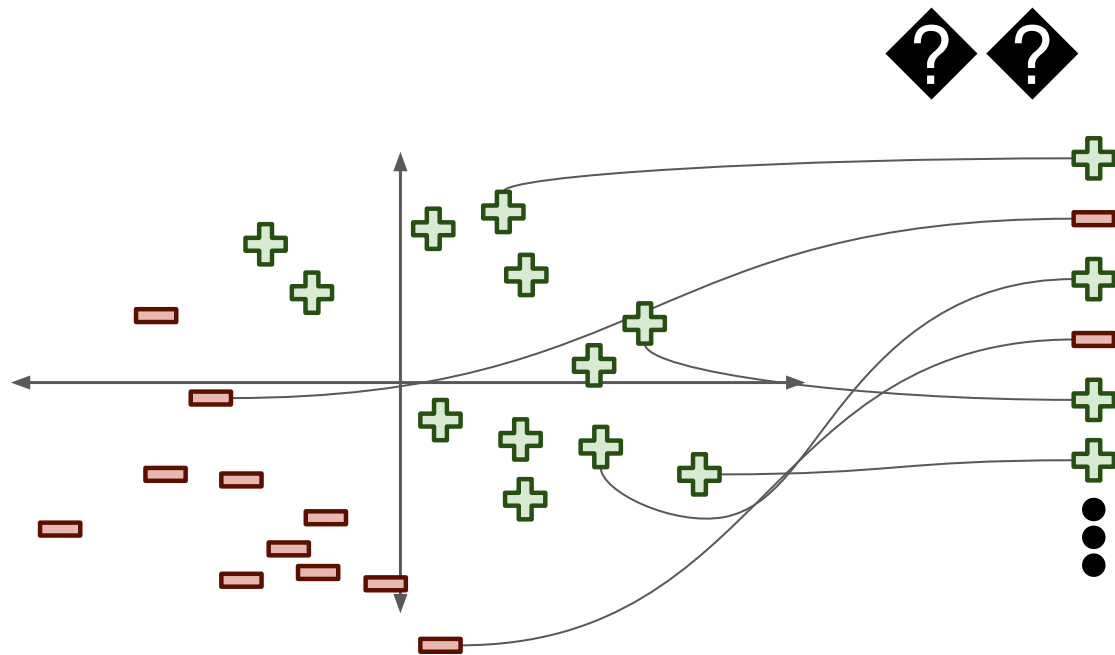
What if we *just* care about the observed training data?



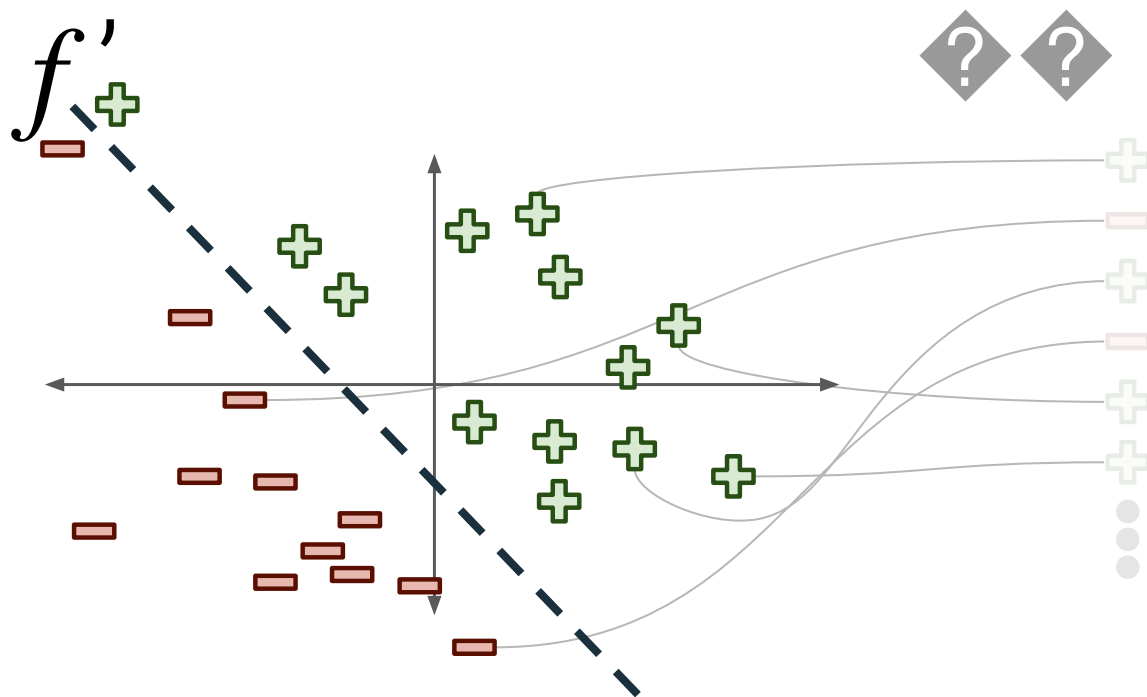
My simple solution: Store all the answers.



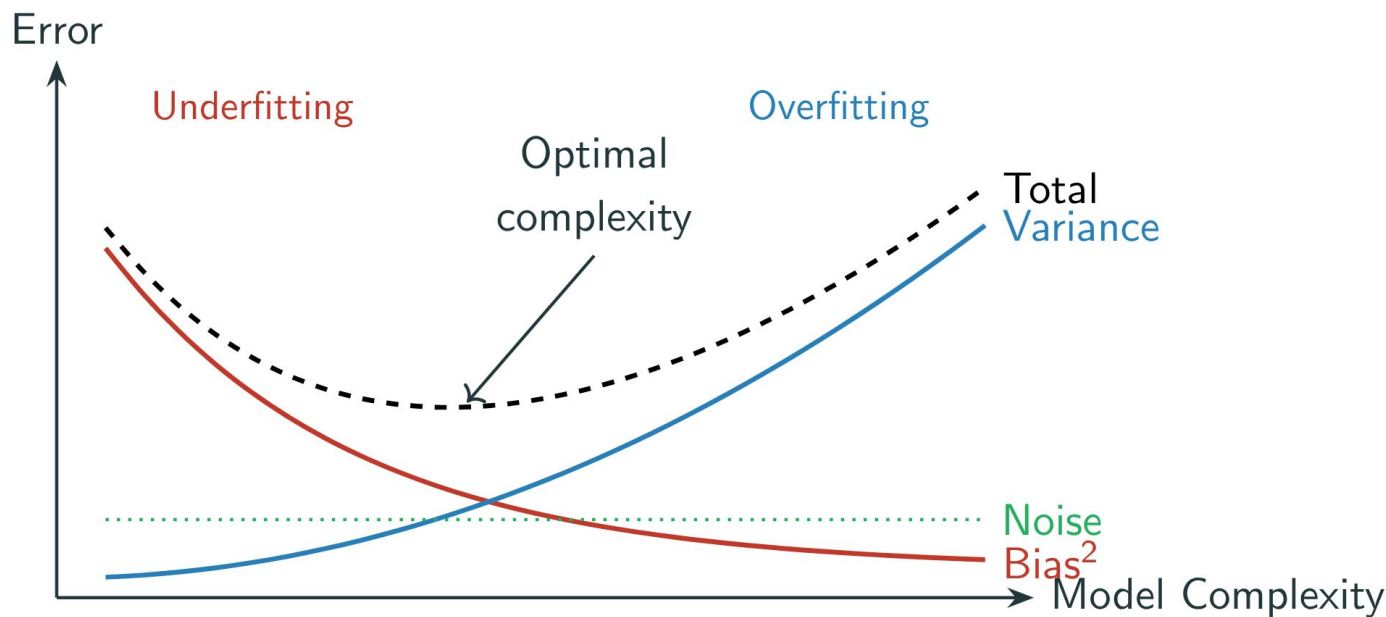
On a different dataset



Maybe I can use less information to store a “simpler” model.

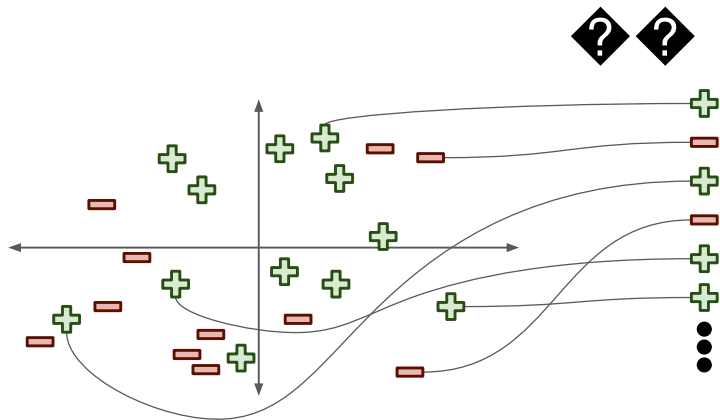
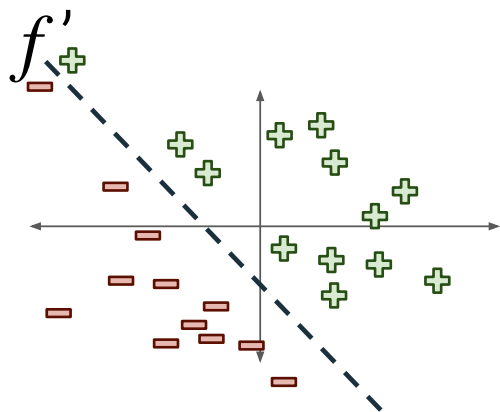


Model “Capacity” or “Complexity” reflects the model’s ability to fit complex functions. As capacity increases, bias decreases and variance increases.



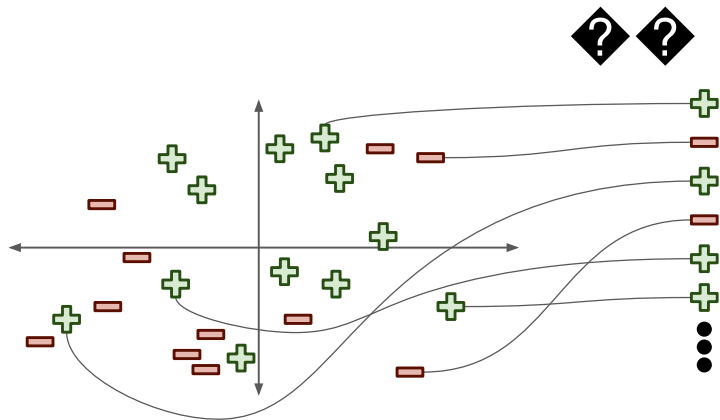
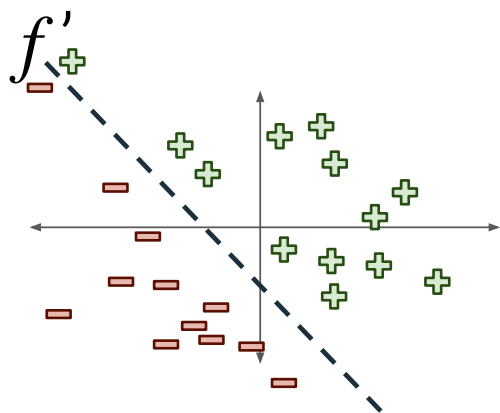
Two takeaways:

1. Memorizing the data is a “perfect” solution (*that I don't want the model to use*).



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1. Memorizing the data is a “perfect” solution (*that I don't want the model to use*).
2. Maybe if I make sure my model is sufficiently simple (e.g., low parameters), it won't be able to use this solution?

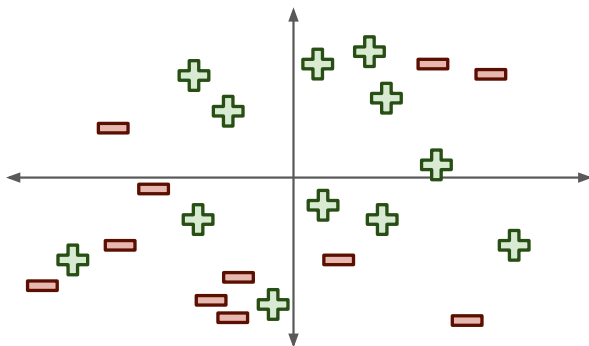


Regularization is the process of constraining the model's complexity to limit memorization!

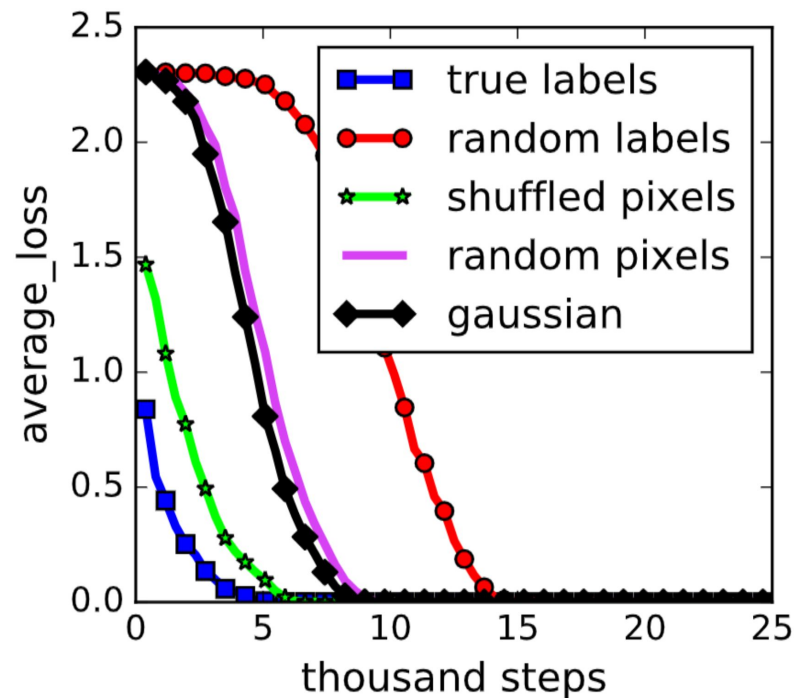
The Generalization Hypothesis

Option 1: Modern Deep Learning methods are so regularized and training datasets so large that it is not feasible to memorize all the training data in the model.

Option 2: Modern Deep Learning methods are sufficiently complex and training datasets sufficiently small that they can memorize the training data, they just... don't.



Option 2 is correct.



(a) learning curves

Why not?

If my model can
memorize the data,
why doesn't it?

If a perfect optima exists, why won't the computer use it, when all you've told it to do is optimize the function?



This is **THE** question for the theory of AI.

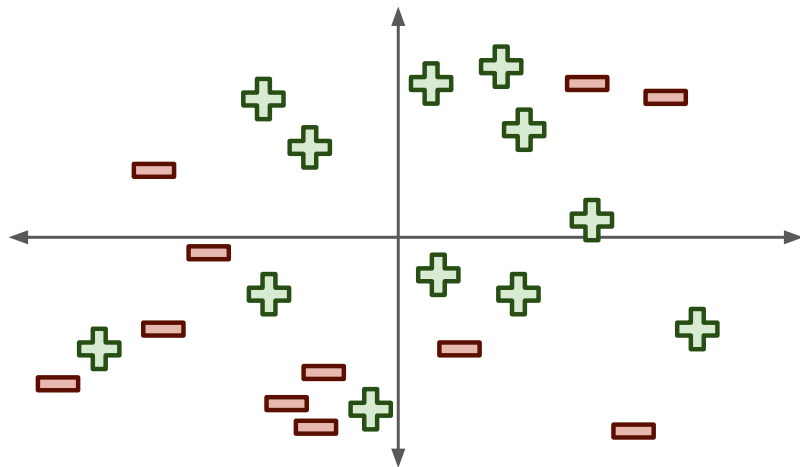
Without answering it, we don't know why AI works at all!

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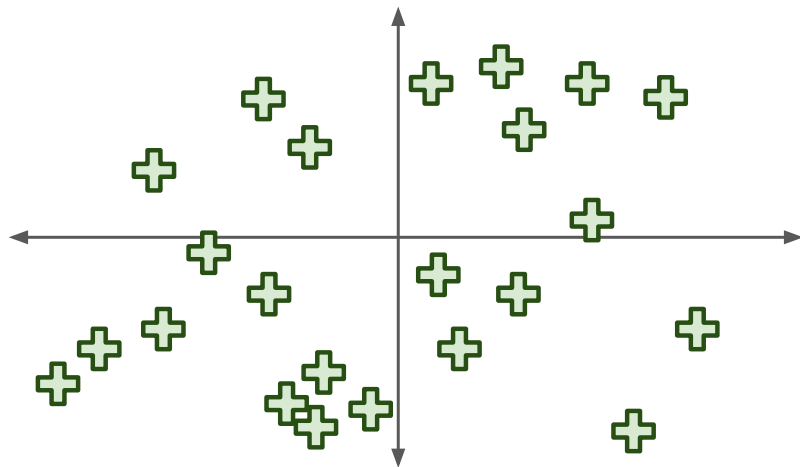
Key Idea:

It isn't about how many *parameters* it takes to encode a model that generalizes, it is about how easy it is to learn a model that generalizes!

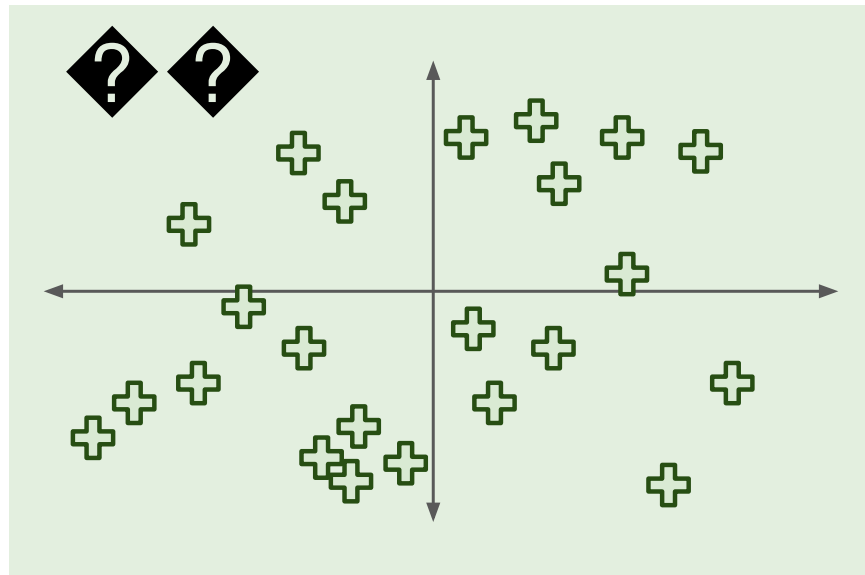
Example: Sometimes Memorization is Right!



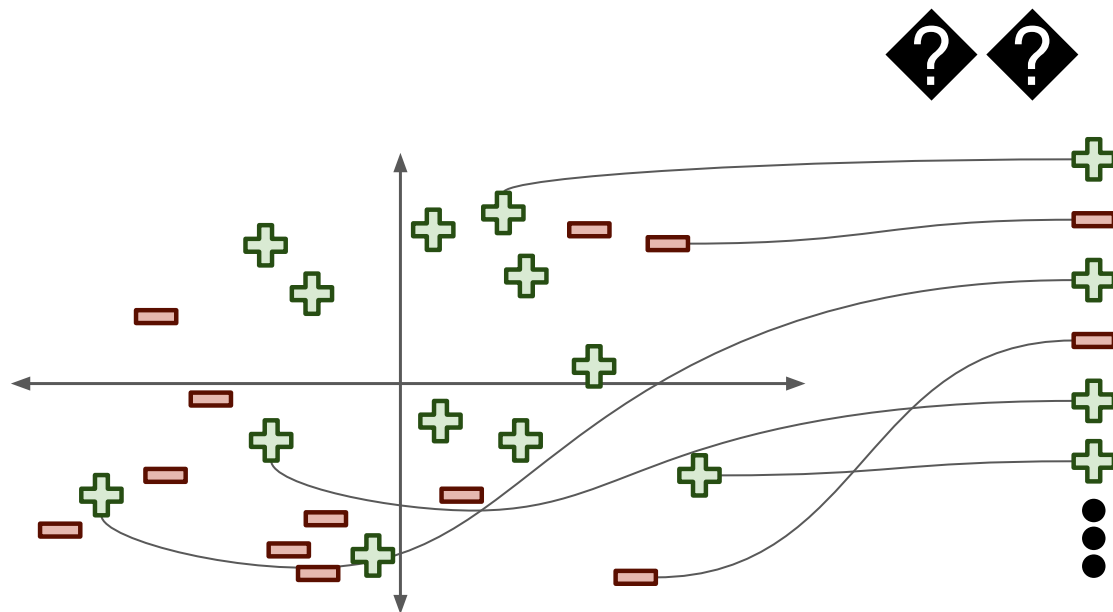
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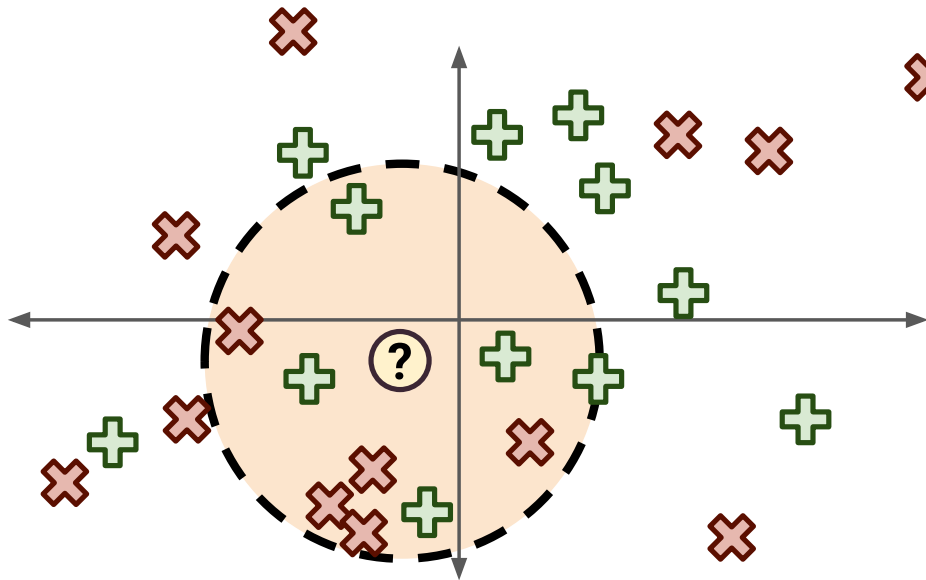
Example: Sometimes Memorization is Right!



Memorizing seems bad...



But interpolation (e.g., averaging in neighborhoods) can be good?



Interpolation (via a nearest neighbor model) is, in the limit, a *perfect* model!

If your dataset size grows towards infinity, and the number of points you include in your neighborhood grows towards infinity, and the ratio of the number of points in your neighborhood divided by your dataset size goes to zero, then a nearest neighbor model converges to *the* Bayes optimal model.

$$\lim_{n \rightarrow \infty, k \rightarrow \infty, \frac{k}{n} \rightarrow 0} k\text{-nn} = \text{Bayes Optimal}$$

Inductive Bias:

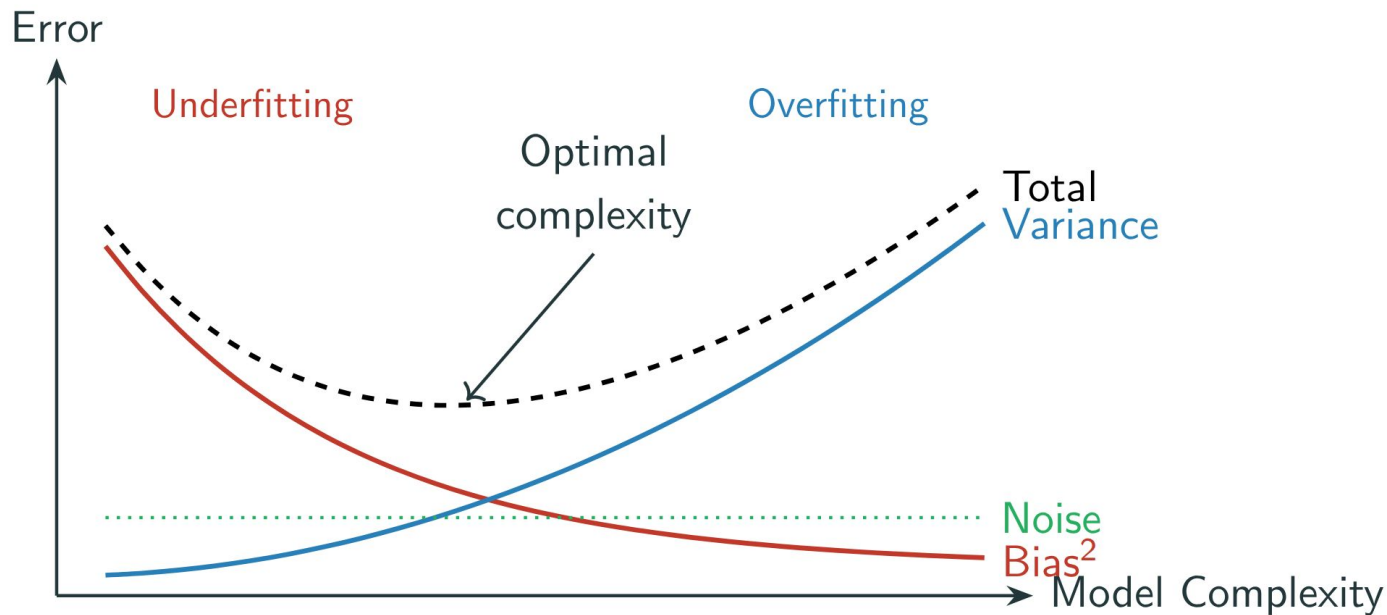
“How easy it is to learn
the right model”

Native and Introduced Inductive Bias

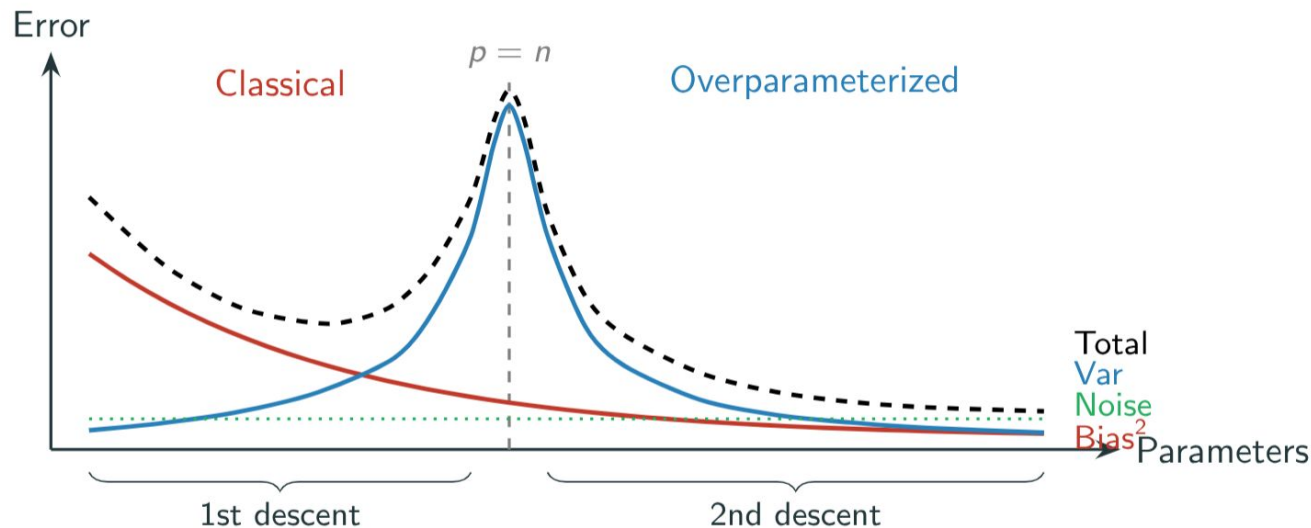
There are (to some extent) four (non-independent) kinds of inductive bias:

1. Inductive bias introduced by the fundamental learning process employed in nearly all modern ML (stochastic gradient descent over likelihood like losses)
e.g., interpolation, double descent.
2. Inductive bias introduced by the dataset, captured in the model via modeling choices
e.g., manifold hypothesis, areas of support and directions of variance in the data.*
3. Inductive bias captured through direct, unavoidable constraints in the model class.
*e.g., neural network architectures**
4. Inductive bias captured through modifications to your training objective to enforce indirect, soft constraints in the model class.
e.g., structure-inducing pre-training

Is bias-variance tradeoff wrong?

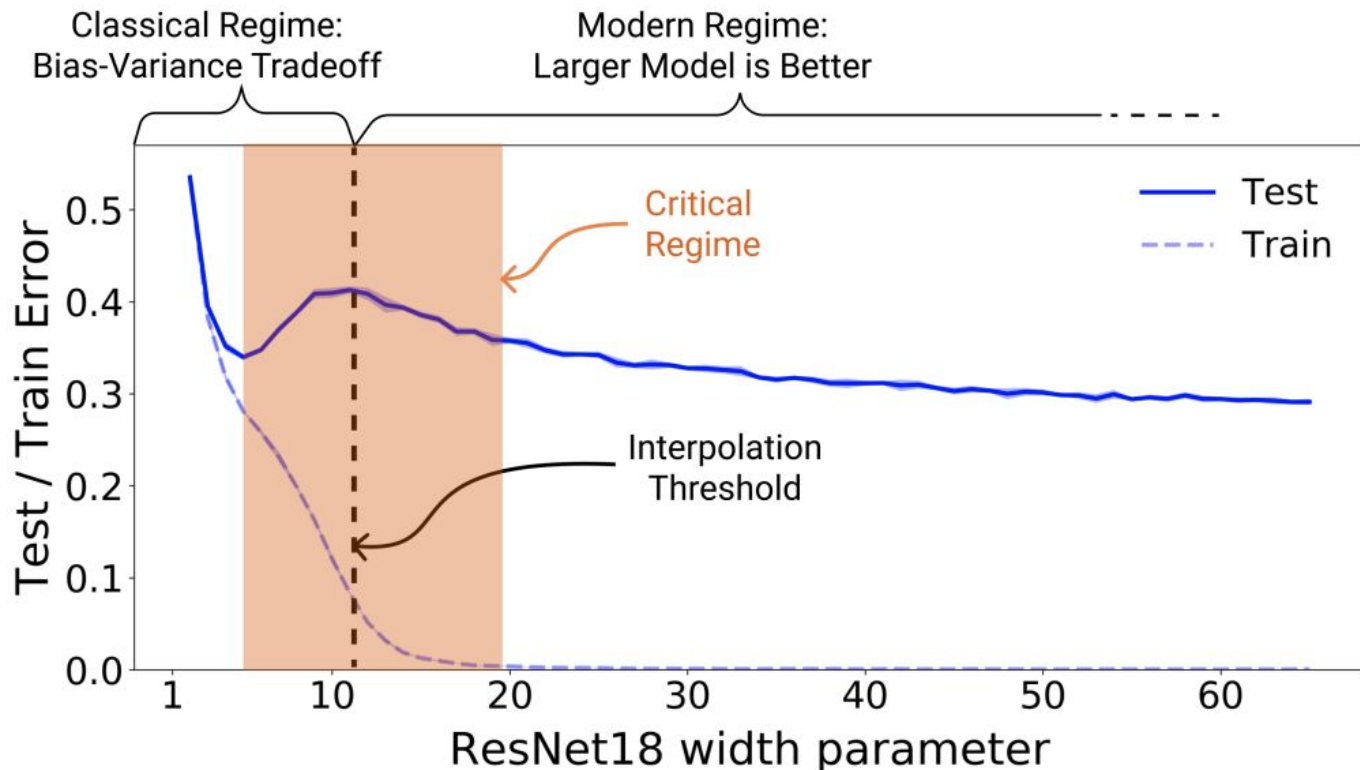


Is bias-variance tradeoff wrong? No -- it just may be incomplete.



At $p = n$: model interpolates training data. Beyond: more parameters $\Rightarrow$ lower test error!

A more precise, realistic picture (we only measure error).



Connections

This question -- *Why does AI work?* -- connects to every other aspect of math we've studied in this course.

- *Linear algebra*: The spectral decomposition (eigenspace) of the data through the model kernel dictates interpolation behavior.
- *Probability & Statistics*: Data distributions induce geometry through directions of variance and density.
- *Calculus & Optimization*: Gradient descent powers everything and explicitly introduces inductive bias when overparametrized. Specification or alteration of your objective function gives a controllable lever to adjust the learning process.

How does this help
you?

Doing Machine Learning, Right

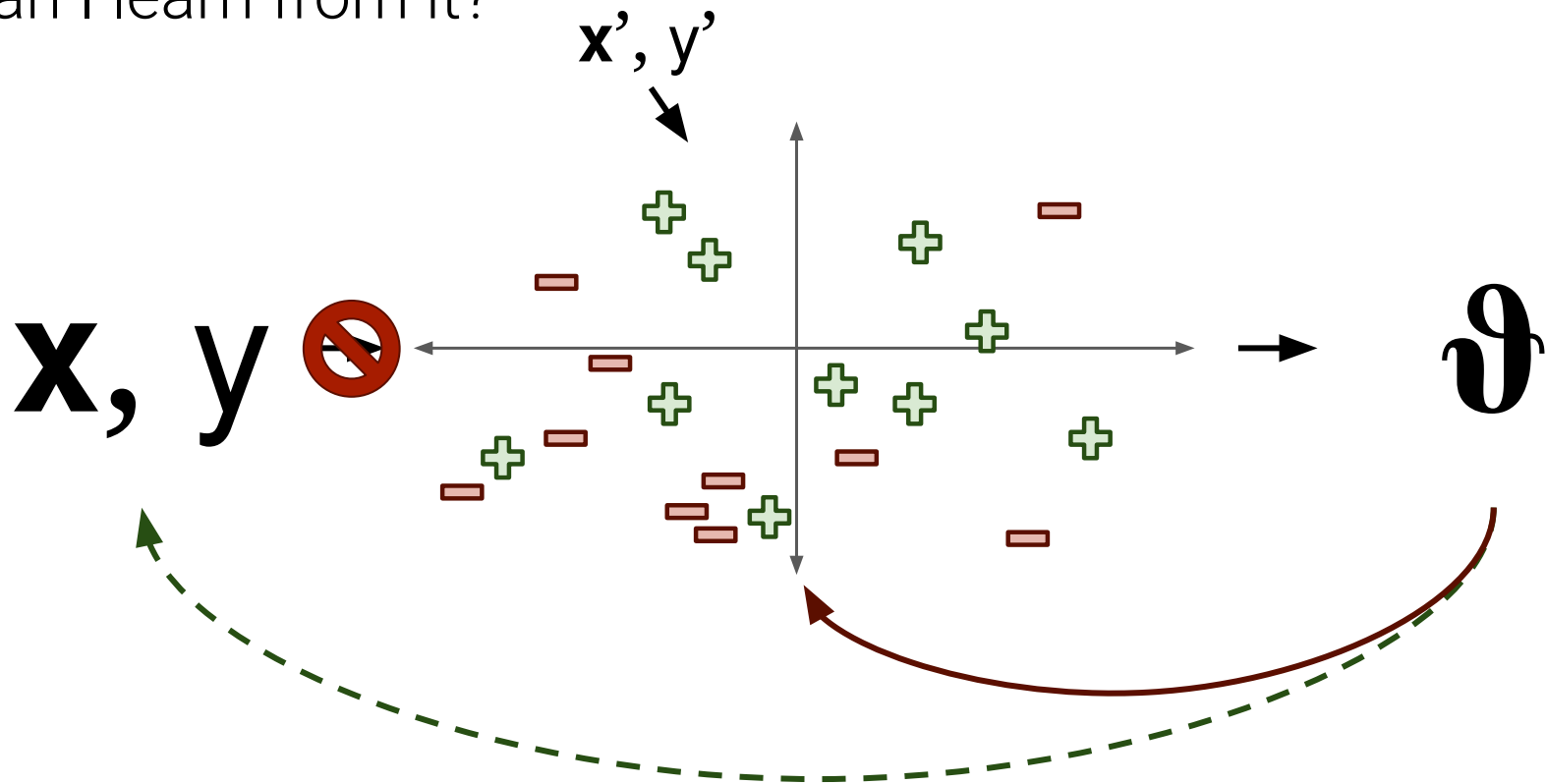
1. Ensure you do your experiments right
Control for and measure the right things, run negative controls, baselines, etc.
2. Understand your problem.
What “invariances” hold for your task? Why will your model class respect them?
3. Understand your data.
What is the structure in your data? Why is your model class appropriate?
4. Understand your model.
*What **can't** your model express? How fast should it converge on simple data?
How do you know it is working?*
5. Understand your learning algorithm.
Does your loss permit trivial optimization? Why should your algorithm work?

What can't (anyone)
do?

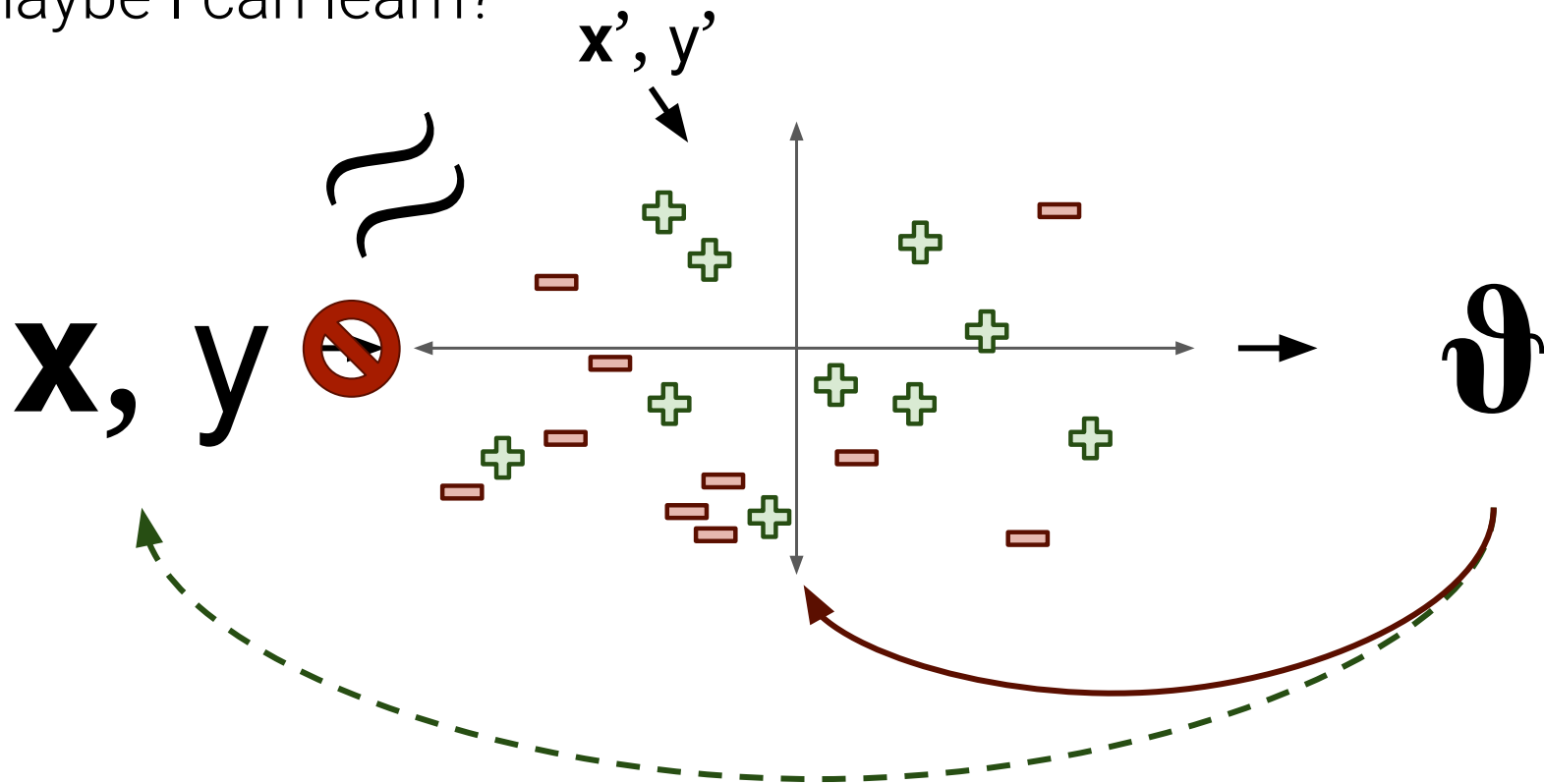
Unsolvable Problems?

1. Distribution shift (without assumptions).
I can't sample $\mathbf{x}, \mathbf{y}$, but I only care about performance on $\mathbf{x}, \mathbf{y}$. Can AI solve it?
2. Explainability (without very precise definitions)
But can you tell me why *f* really works?
3. Fairness (on non-trivial problems)
Can a model satisfy both separation and sufficiency at once?

Distribution shift: If you don't let me sample from $\mathbf{x}, \mathbf{y}$, how can I learn from it?



Distribution shift: If you *almost* let me sample from $\mathbf{x}, y$, maybe I can learn?



Explainability: You (wanna) catch them all
But you can't. Probably.

- Faithfulness: explanation reflects the true decision mechanism
- Simplicity / sparsity: explanation is compact / human-interpretable
- Stability: small input changes $\rightarrow$ small explanation changes
- Completeness: explanation accounts for all of the model's behavior
- Model-agnosticism: explanation applies across architectures
- Counterfactual validity: explanation aligns with what would change the outcome

Fairness: You (wanna) catch them all
But you can't. Definitely.

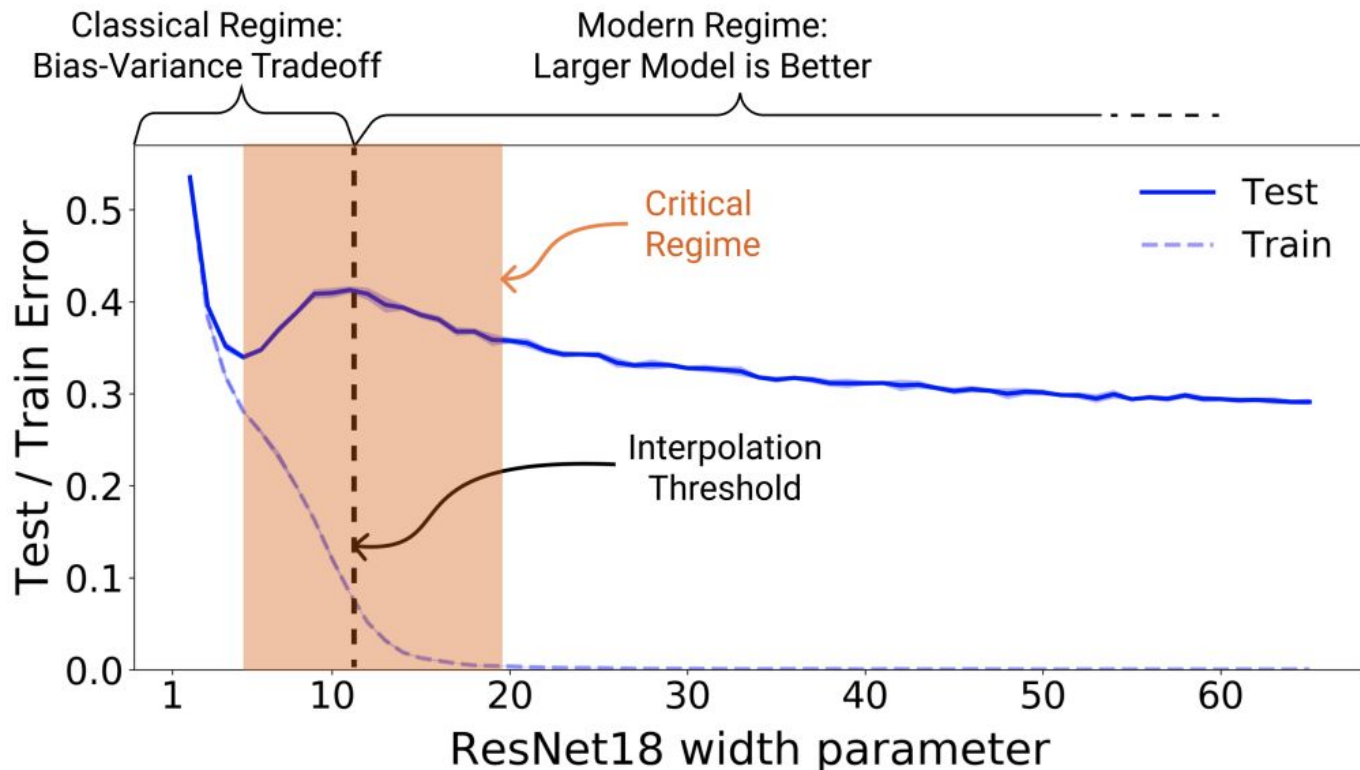
- Suppose Y is my label, $\hat{Y}$ is my prediction, and A is a group attribute (e.g., socioeconomic status), all related in a joint random variable.
- Separation: $\hat{Y} \perp A \mid Y$
- Sufficiency: $Y \perp A \mid \hat{Y}$

It is not mathematically possible aside from on degenerate random variables to satisfy both separation and sufficiency.

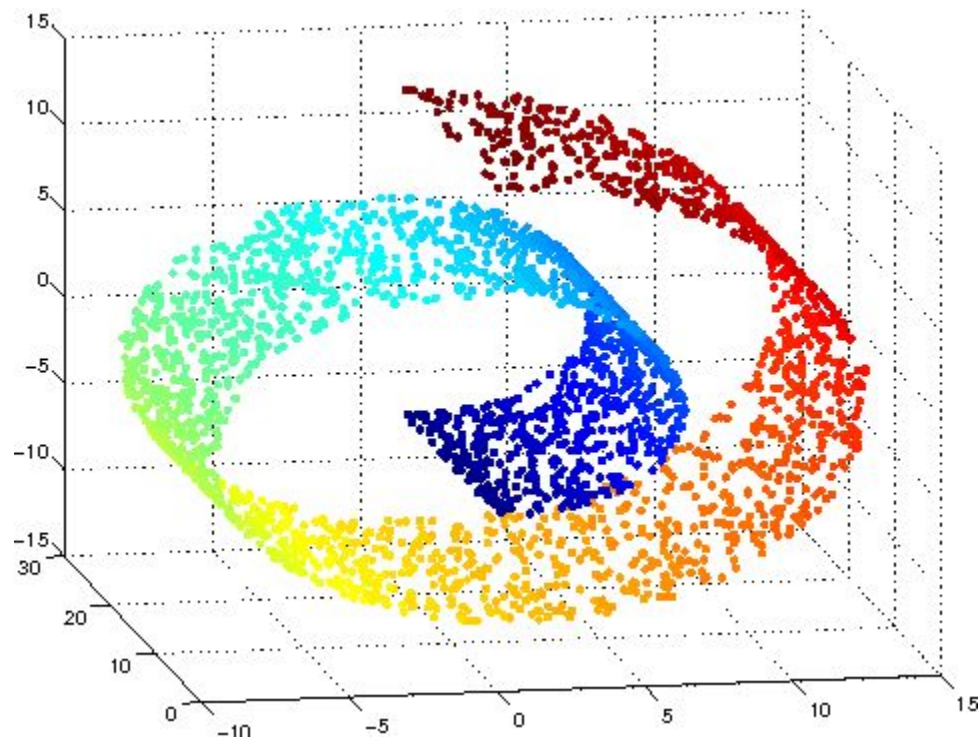
Next time:

What about real problems -- different kinds of losses, optimization functions, etc. How can we operationalize different problems?

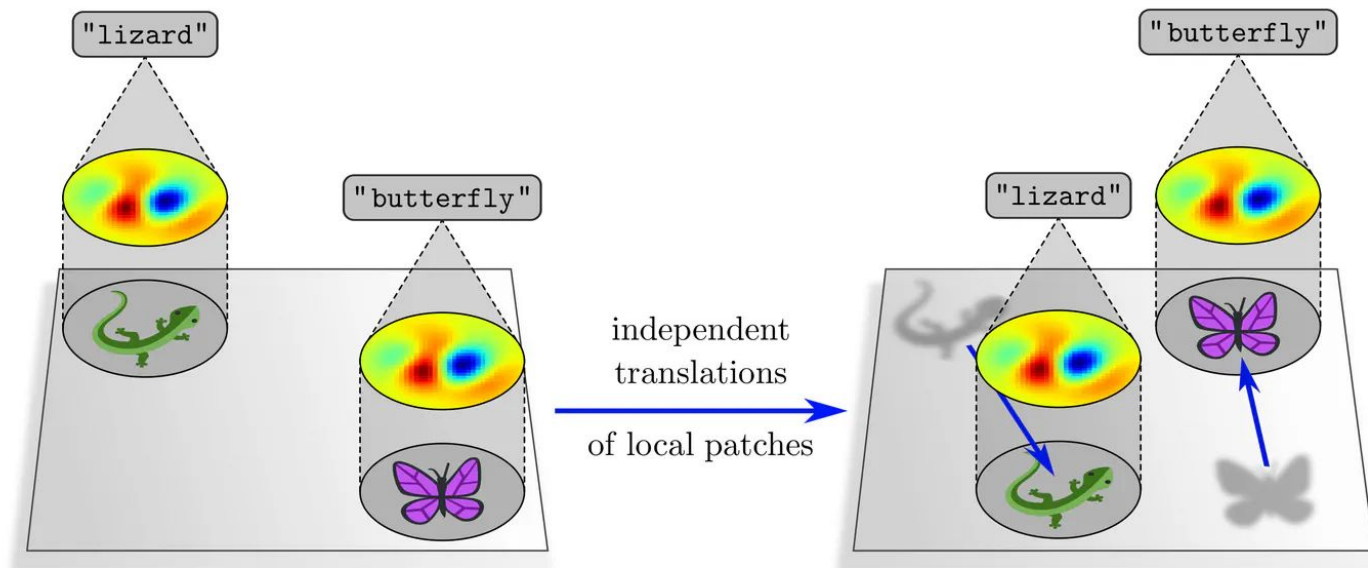
Inductive Bias “Quick Hits”: Interpolation / Double Descent



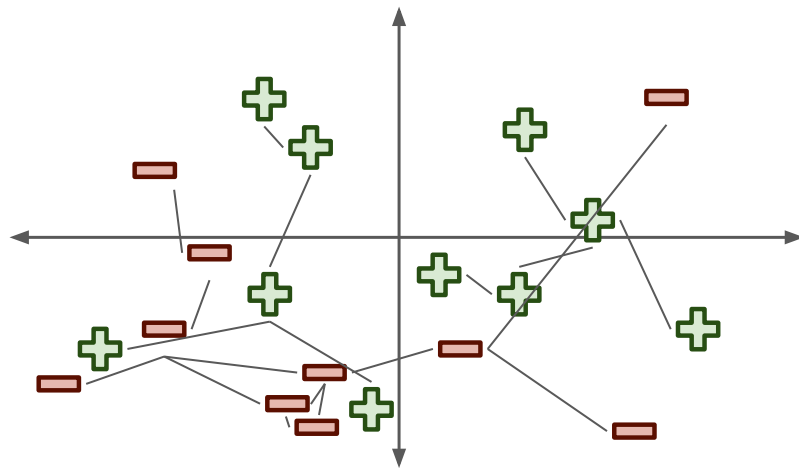
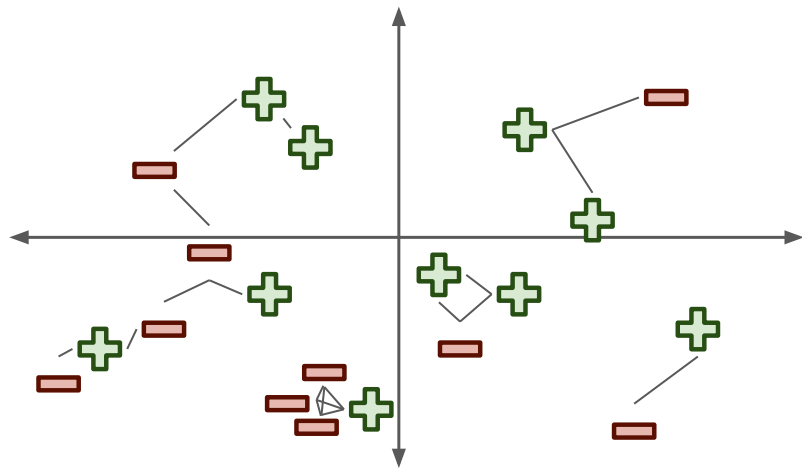
Inductive Bias “Quick Hits”: Data is a manifold



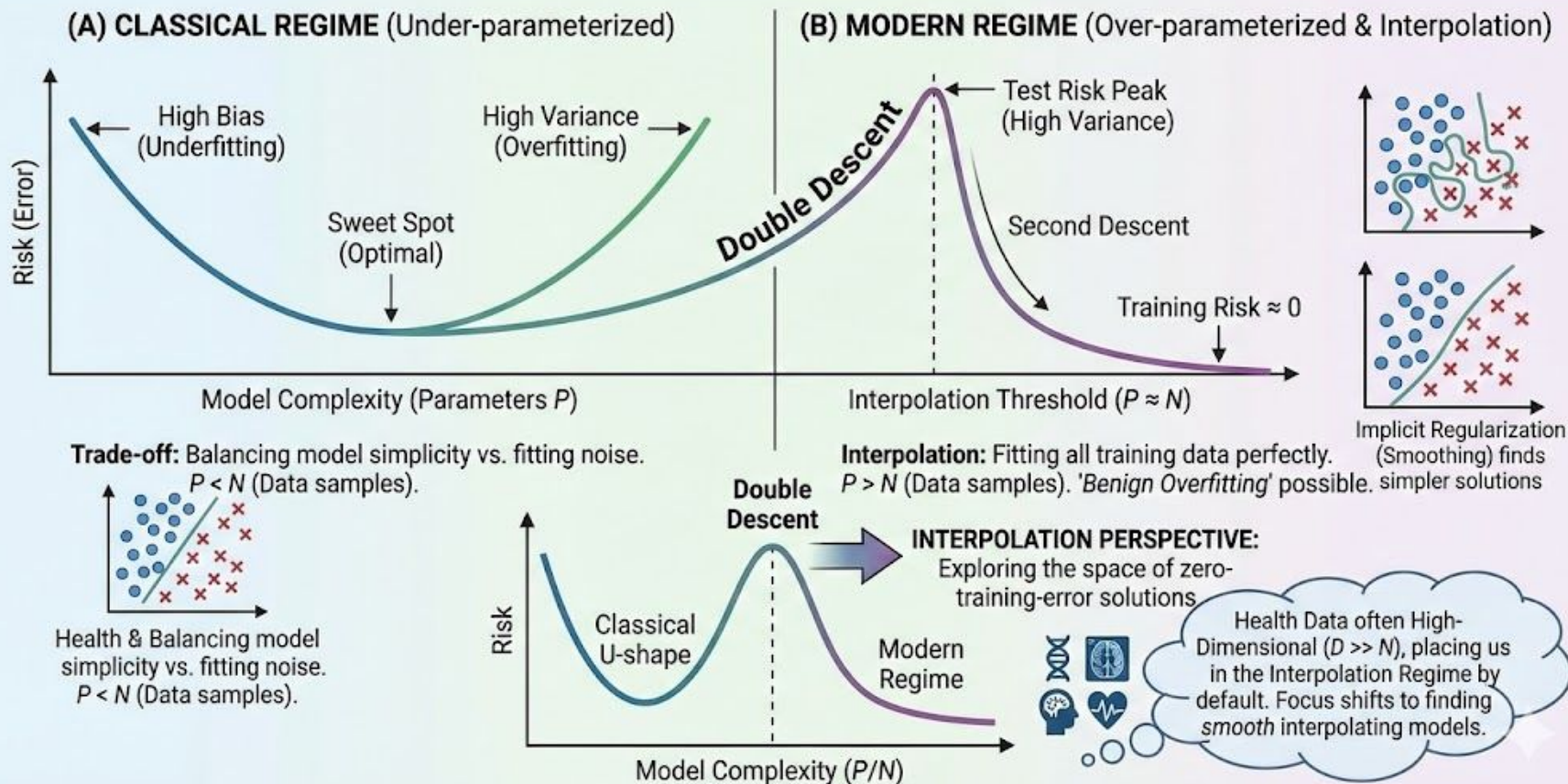
Inductive Bias “Quick Hits”: CNNs are translation invariant



Inductive Bias “Quick Hits”: Learning a geometry captures geometry!



DOUBLE DESCENT & THE INTERPOLATION PERSPECTIVE IN HEALTH ML

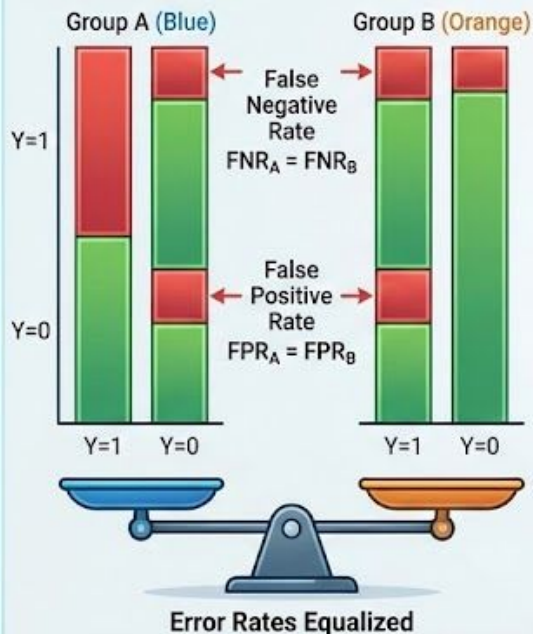


THE FAIRNESS TRADE-OFF: SEPARATION VS. SUFFICIENCY (IMPOSSIBILITY THEOREM)

1. SEPARATION

(Equalized Error Rates)

Goal: Prediction (R) $\perp$ Attribute (A) | True Label (Y)



CONFLICT ZONE

IMPOSSIBILITY THEOREM

$R \perp A | Y$

$\neq$

$Y \perp A | R$

Cannot be simultaneously satisfied

Reality Check: Unless base rates are equal ($P(Y|A)=P(Y|B)$) or the model is perfect ($Y=R$), you must **CHOOSE ONE**. They are mathematically incompatible goals.

2. SUFFICIENCY

(Equalized Predictive Value)

Goal: True Label (Y) $\perp$ Attribute (A) | Prediction (R)

